



Plastic Debris in the Chattogram Coast and Karnaphuli Estuary, Bangladesh: Sources, Pathways, Hotspots, and Policy Gaps



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ABSTRACT

Plastic debris is a critical environmental and governance challenge in the Karnaphuli estuary. Using surveys and spatial mapping, this study assessed plastic sources, pathways, and impacts. Results show 78% of respondents consider plastic pollution a worsening issue. Markets and households are dominant sources, while drains and stormwater are primary pathways, especially during rainfall. Fieldwork identified eight accumulation hotspots. This pollution severely impacts aquatic life, fish productivity, and local livelihoods. These issues stem from systemic

waste-management failures, including inadequate infrastructure, open dumping, and weak enforcement. Plastic pollution here is a pathway-specific governance problem requiring integrated land-to-sea management and stronger accountability.

Keywords: Plastic Debris, Transport Pathways, Accumulation Hotspots, Coastal-Estuarine System, Governance Gaps, Waste Management Infrastructure, Spatial Analysis, Bay of Bengal.

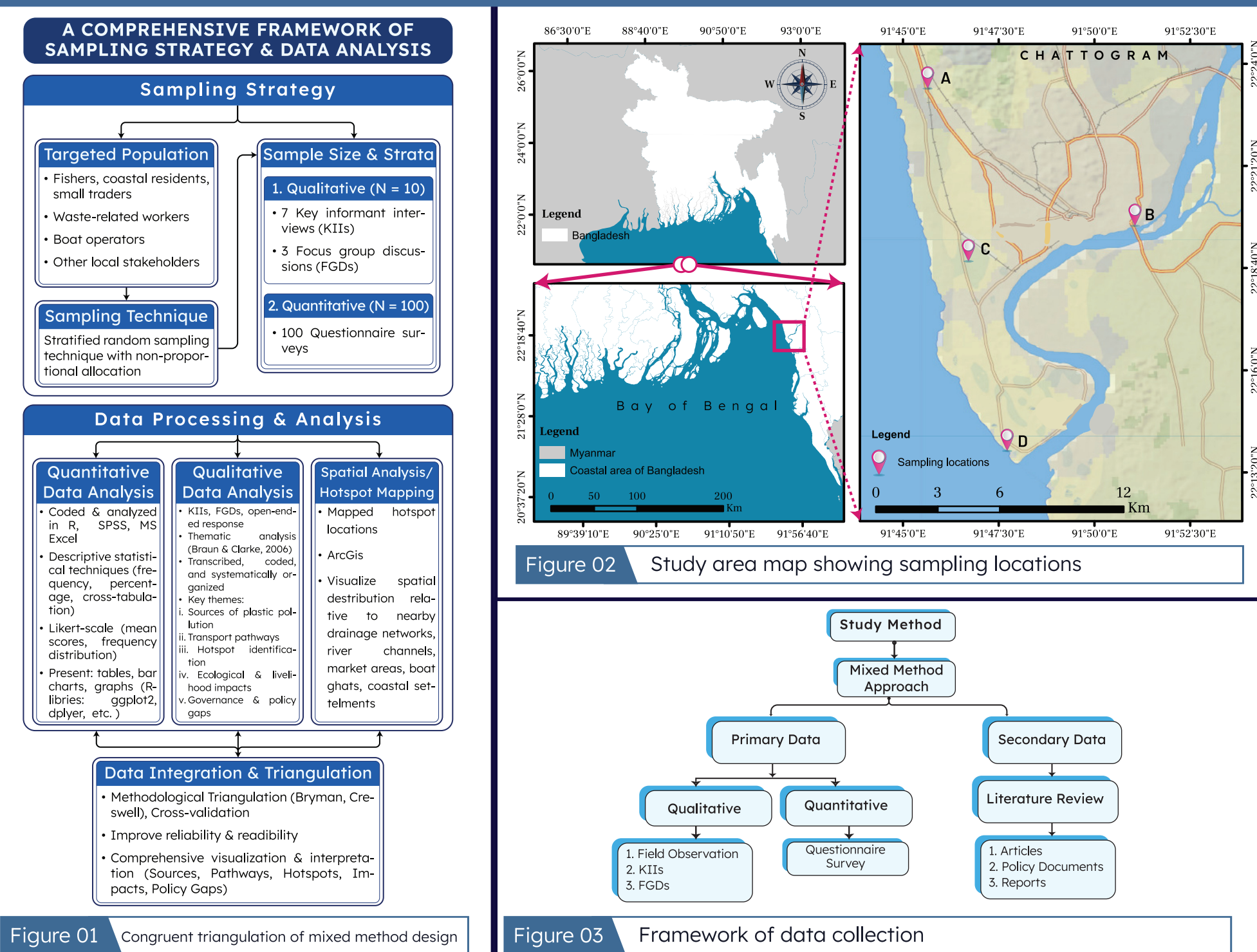
INTRODUCTION

Marine plastic debris is a critical global crisis; of the 9 billion tons produced since 1950, only 9% has been recycled. In Bangladesh, this crisis is amplified along the Chattogram coast and Karnaphuli estuary. This vital coastal system receives 2,100–2,400 tons/day of municipal solid waste, with >230 tons of single-use polymers flushed directly into the Bay of Bengal through 57 urban drainage canals. This escalating contamination severely threatens local fisheries and sensitive ecosystems, including St. Martin's coral habitats, demanding targeted, evidence-based governance

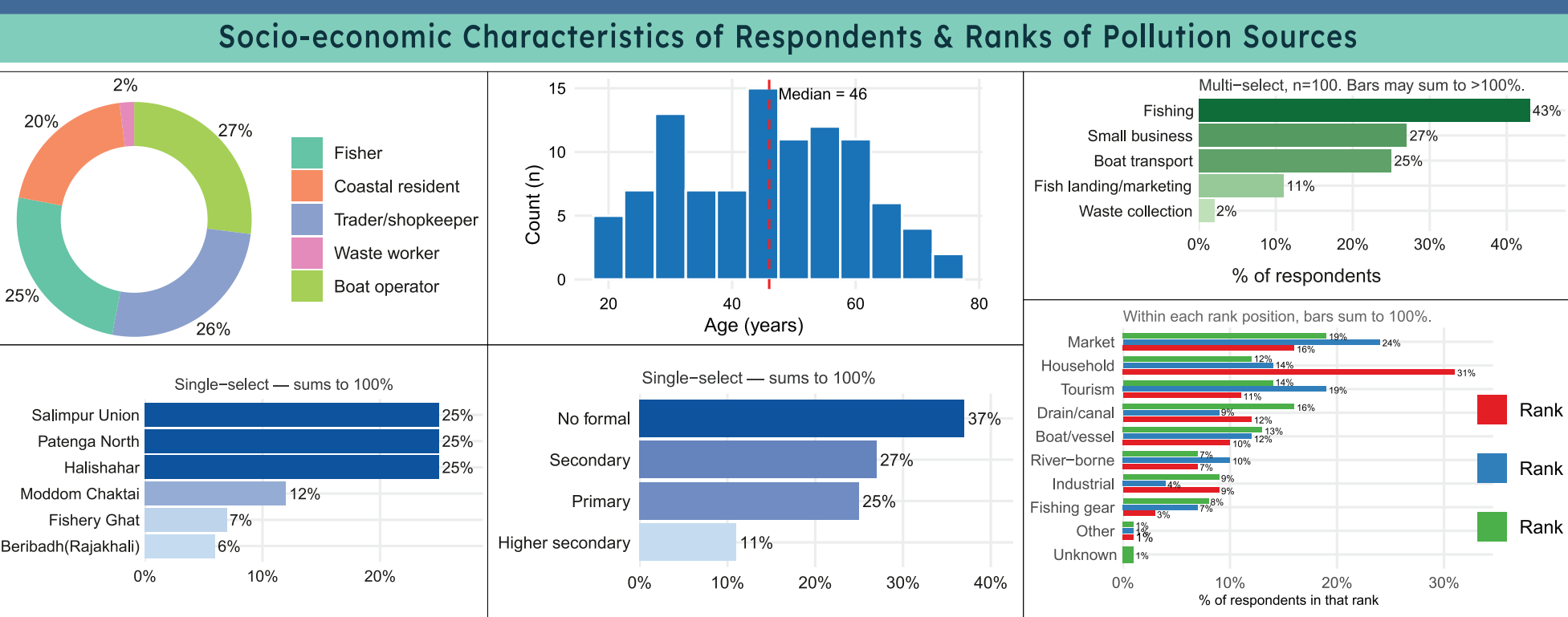
OBJECTIVES

- To identify the major sources and transport pathways through which plastic debris enters coastal and marine environments.
- To map key plastic concentration zones using stakeholder-driven data and field observations.
- To assess the reported ecological and livelihood impacts of plastic debris among coastal communities.
- To examine institutional and waste management gaps contributing to plastic discharge into the marine environment.

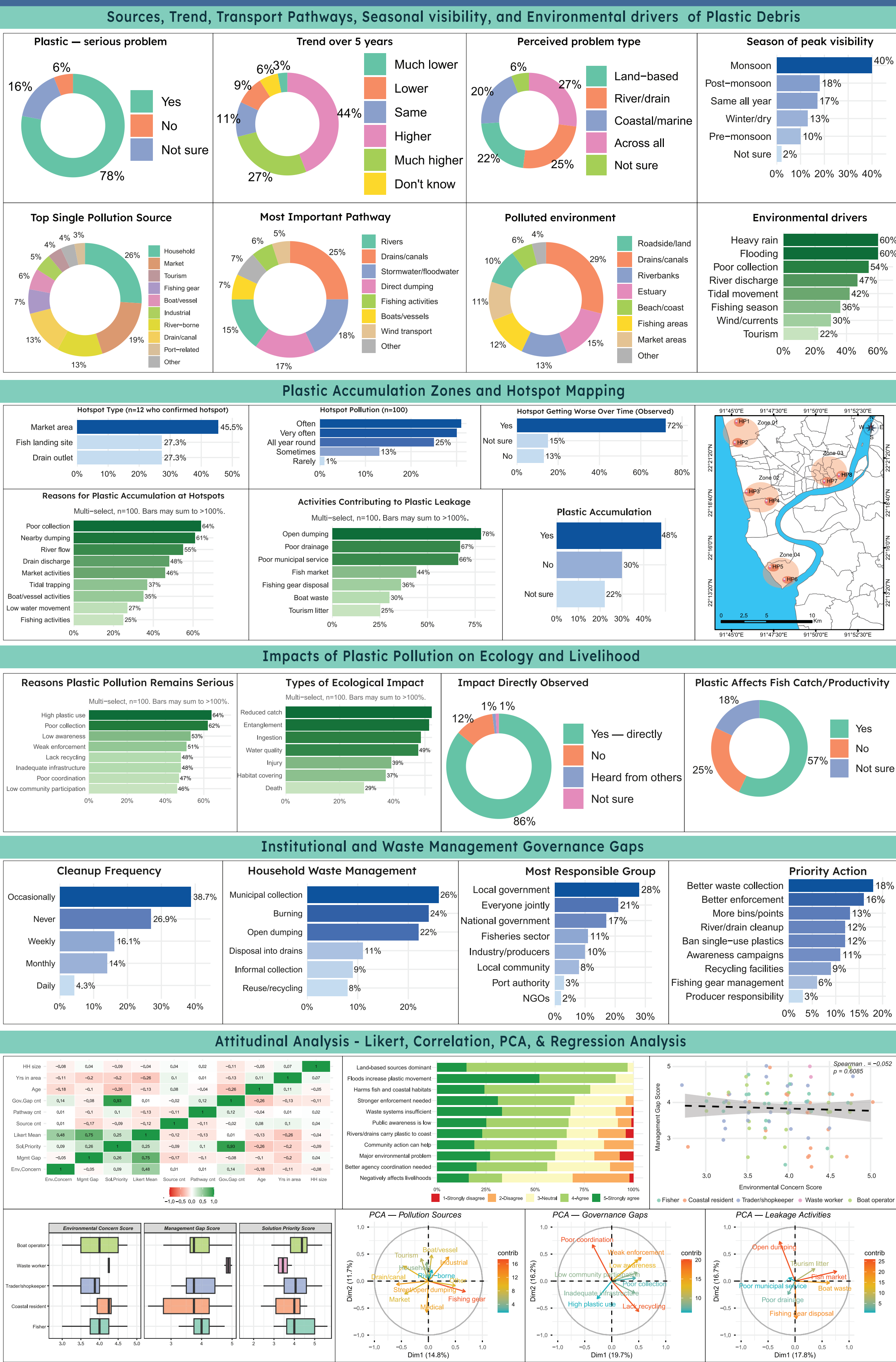
METHODOLOGY



RESULTS



RESULTS CONTINUE



CONCLUSION

Plastic pollution in the Karnaphuli estuary is a systemic governance crisis driven by municipal waste leakage, not just local littering. Stakeholder surveys and field data reveal local market and domestic wastes are the primary sources, heavily funneled through drainage networks, canals, and monsoon floodwaters into coastal ecosystems. This unmanaged transport degrades aquatic habitats and severely impacts fisheries-based livelihoods. Currently, weak enforcement and insufficient infrastructure fail to contain the leakage. Controlling this crisis requires an urgent, pathway-based management framework that establishes clear institutional responsibility, upgrades urban waste collection, controls drainage pathways, and introduces regular hotspot monitoring.

REFERENCES

